VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of the UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

M.Tech Course: Machine Learning Lab (MACSE504)

CASE STUDY BASED MINI-PROJECT

EVALUATION SCHEME & RUBRIC (TOTAL MARKS: 30)

Each case study mini-project question is structured into three mandatory parts:

- **Part A: Exploratory Data Analysis & Preprocessing** — [10 Marks] **Deadline: 5th October 5, 2026**
  (Data cleaning, missing value handling, statistical profiling, distribution analysis, correlation analysis)
- **Part B: Feature Engineering & Model Fitting** — [10 Marks] **Deadline: 12th October 5, 2026**
  (Feature extraction/selection, encoding, baseline and complex model training, cross-validation, hyperparameter tuning)
- **Part C: Model Evaluation, Visualization & Domain Inference** — [10 Marks] **Deadline: 26th October 5, 2026**
  (Performance metrics interpretation, diagnostic visualization, error analysis, practical domain-specific inferences)

***provide an interface to load dataset, training, parameter adjustment and for visualization.

Reg.No	Q.No		Reg.No	Q.No		Reg.No	Q.No
25PHD2036	1		26MML0020	1		26MML0043	1
26MML0001	2		26MML0021	2		26MML0044	2
26MML0002	3		26MML0022	3		26MML0045	3
26MML0003	4		26MML0023	4		26MML0046	4
26MML0004	5		26MML0024	5		26MML0047	5
26MML0005	6		26MML0025	6		26MML0049	6
26MML0006	7		26MML0026	7		26MML0050	7
26MML0007	8		26MML0028	8		26MML0051	8
26MML0008	9		26MML0030	9		26MML0052	9
26MML0009	10		26MML0031	10		26MML0053	10
26MML0010	1		26MML0032	1		26MML0054	1
26MML0011	2		26MML0033	2		26MML0055	2
26MML0012	3		26MML0034	3		26MML0056	3
26MML0013	4		26MML0035	4		26MML0057	4
26MML0014	5		26MML0036	5		26MML0058	5
26MML0015	6		26MML0037	6		26MML0059	6
26MML0016	7		26MML0038	7		26MML0060	7
26MML0017	8		26MML0039	8		26MML0061	8
26MML0018	9		26MML0041	9		26MML0062	9
26MML0019	10		26MML0042	10		26MML0063	10
26MML0064	1		26MML0066	3		26MML0068	5
26MML0065	2		26MML0067	4			

Question 1: Predictive Churn Risk Modeling for Telecom Subscription Services

Dataset Reference: Telco Customer Churn Dataset

(https://www.kaggle.com/datasets/blastchar/telco-customer-churn**)**

Industry Case Scenario: A telecommunications enterprise wants to reduce customer attrition by identifying subscriber churn risk early. The dataset contains customer demographic profiles, subscribed services, account details, and monthly charges.

Part A: Exploratory Data Analysis & Preprocessing

1. Load the dataset, parse 'TotalCharges' into numeric format, and handle missing values.
2. Analyze customer churn proportions across contract types (Month-to-month, One year, Two year) and internet service types.
3. Generate tenure vs. monthly charges distribution plots and correlation matrices for continuous numerical attributes.

Part B: Feature Engineering & Model Fitting

1. Encode categorical attributes using One-Hot Encoding and Binary Encoding where-ever applicable.
2. Construct Logistic Regression/ K-Nearest Neighbors (KNN) and Random Forest classification models.
3. Apply SMOTE or class-weight adjustment to handle target imbalance and perform 5-fold Stratified Cross-Validation.

Part C: Model Evaluation, Visualization & Business Inference

1. Evaluate model performance using Precision, Recall, F1-Score, and ROC-AUC curves.
2. Plot Confusion Matrices and Feature Importance rankings to identify key churn drivers.
3. Derive business inferences on retention strategies for customers with short tenure and high monthly charges.

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Question 2: Supervised Diabetes Risk Prediction and Biomarker Screening

Dataset Reference: Pima Indians Diabetes Database

(https://www.kaggle.com/datasets/kumargh/pimaindiansdiabetescsv)

Industry Case Scenario: A healthcare provider requires a clinical decision support tool to predict whether a patient has diabetes based on diagnostic medical measurements (Glucose, Blood Pressure, Insulin, BMI, Age).

Part A: Exploratory Data Analysis & Preprocessing

1. Inspect diagnostic measurements and detect biologically impossible zero values in Glucose, Blood Pressure, and BMI.
2. Impute non-zero missing indicators using median values conditioned on target outcome classes.
3. Analyze feature correlations and generate pair plots comparing diabetic vs. non-diabetic patient profiles.

Part B: Feature Engineering & Model Fitting

1. Standardize features using StandardScaler to ensure distance-based models operate on equal scales.
2. Fit Support Vector Machines (SVM) with linear and RBF kernels alongside a K-Nearest Neighbors (KNN) classifier.
3. Perform Grid Search CV to optimize hyperparameter C, gamma, and n_neighbors.

Part C: Model Evaluation, Visualization & Clinical Inference

1. Calculate Sensitivity (Recall), Specificity, Accuracy, and ROC-AUC for all fitted classifiers.
2. Plot ROC Curves and decision boundaries for the top two diagnostic features.
3. Provide clinical inferences regarding the threshold levels of Glucose and BMI that trigger high diagnostic risk.

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Question 3: Used Car Valuation & Elasticity Analysis using Regularized Regression

Dataset Reference: Vehicle Dataset from CarDekho

(https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho)

Industry Case Scenario: An online automotive marketplace seeks an automated valuation model to estimate used car resale prices based on vehicle age, mileage driven, fuel type, transmission type, and dealer type.

Part A: Exploratory Data Analysis & Preprocessing

1. Calculate vehicle age from manufacturing year and clean string values in mileage and engine capacity features.
2. Analyze price distribution skewness and apply log-transformation to the target variable 'Selling_Price'.
3. Visualize price trends against kilometers driven and fuel types using scatter plots and boxplots.

Part B: Feature Engineering & Model Fitting

1. Create interaction terms (e.g., age-to-mileage ratio) and perform One-Hot Encoding on categorical features.
2. Build Multiple Linear Regression, Ridge (L2), and Lasso (L1) regularized regression models.
3. Perform cross-validation to select optimal regularization hyperparameter (alpha).

Part C: Model Evaluation, Visualization & Market Inference

1. Measure performance using MAE, RMSE, and R-squared metrics on unseen test data.
2. Plot Residual Graphs (Residuals vs Predicted) and Feature Coefficient magnitudes.
3. Provide commercial inferences on vehicle depreciation rates and the resale value premium of diesel vs. petrol cars.

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Question 4: Automotive Fleet Profiling & Efficiency Clustering

Dataset Reference: Auto MPG Dataset

(https://archive.ics.uci.edu/ml/datasets/Auto+MPG)

Industry Case Scenario: An automotive engineering group aims to categorize vehicle models into structural efficiency classes based on engine displacement, horsepower, weight, acceleration, and fuel economy (MPG).

Part A: Exploratory Data Analysis & Preprocessing

1. Identify missing values in 'horsepower' and impute using median values by cylinder group.
2. Analyze distributions and pair correlations between vehicle weight, horsepower, and fuel efficiency.
3. Standardize continuous attributes using Z-score scaling to eliminate feature scale dominance.

Part B: Feature Engineering & Clustering Implementation

1. Implement K-Means Clustering and Hierarchical Agglomerative Clustering (Ward's linkage).
2. Determine the optimal number of clusters (K) using the Elbow Method and Silhouette Coefficient across K=2 to 6.
3. Plot the Hierarchical Clustering Dendrogram and extract cluster assignments.

Part C: Model Evaluation, Visualization & Engineering Inference

1. Compute Silhouette Score and Davies-Bouldin Index for evaluation.
2. Visualize clusters in 2D space using Principal Component Analysis (PCA) projection.
3. Interpret cluster centroids to define vehicle archetypes (e.g., Heavy Muscle, Fuel-Efficient Compact).

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Question 5: E-Commerce Transaction Pattern Mining & Cross-Selling Analysis

Dataset Reference: Online Retail Transaction Dataset

(https://archive.ics.uci.edu/ml/datasets/Online+Retail)

Industry Case Scenario: A digital retail supermarket wants to analyze transactional purchase patterns from global orders to discover associated item purchases and design automated recommendation widgets.

Part A: Exploratory Data Analysis & Preprocessing

1. Filter out canceled invoices (InvoiceNo starting with 'C') and postage-related non-product items.
2. Aggregate transactions into a customer-invoice binary transaction matrix.
3. Plot top 15 most frequently ordered products and analyze transaction basket sizes.

Part B: Algorithm Implementation & Mining

1. Implement the Apriori algorithm and FP-Growth algorithm to extract frequent itemsets.
2. Compare execution times between Apriori and FP-Growth for varying minimum support thresholds (0.01 to 0.05).
3. Generate association rules based on Minimum Confidence and Lift metrics.

Part C: Rule Evaluation, Visualization & Commercial Inference

1. Filter rules using Minimum Support, Confidence (>0.6), and Lift (>2.0).
2. Construct scatter plots (Support vs Confidence colored by Lift) and network graph diagrams of top rules.
3. Recommend specific product bundling combinations and digital store page placement strategies.

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Question 6: Semi-Supervised Spam Detection in Low-Resource Email Environments

Dataset Reference: SMS Spam Collection Dataset

(https://archive.ics.uci.edu/ml/datasets/SMS+Spam+Collection)

Industry Case Scenario: A cybersecurity firm requires an email spam filtering classifier. Manual message annotation is expensive, resulting in a scenario where only 10% of training data is labeled and 90% remains unlabeled.

Part A: Exploratory Data Analysis & Text Preprocessing

1. Analyze class distributions (Ham vs. Spam) and text message length distributions.
2. Clean text data: lowercase conversion, punctuation removal, stop-word removal, and TF-IDF feature extraction.
3. Simulate a semi-supervised dataset by masking target labels for 90% of the training instances.

Part B: Semi-Supervised Model Implementation

1. Implement a Self-Training Classifier using Naive Bayes as the base estimator.
2. Implement Label Propagation and Label Spreading graph-based algorithms.
3. Tune pseudo-labeling confidence probability thresholds (e.g., 0.8, 0.9) across self-training iterations.

Part C: Model Evaluation, Visualization & Inference

1. Benchmark semi-supervised models against a baseline supervised model trained only on the 10% labeled set.
2. Plot Macro F1-score and Precision-Recall Curves across iterations.
3. Evaluate the trade-off between pseudo-labeling noise and classification performance gains.

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Question 7: Dimensionality Reduction & Feature Space Compression for Genomic Data

Dataset Reference: Gene Expression Cancer RNA-Seq Dataset

(https://archive.ics.uci.edu/dataset/401/gene+expression+cancer+rna+seq)

Industry Case Scenario: A bioinformatics lab analyzes high-dimensional gene expression data (over 20,000 gene features) to compress dimensions while preserving diagnostic patterns across multiple cancer types.

Part A: Exploratory Data Analysis & Preprocessing

1. Load expression data, check zero-variance features, and remove uninformative gene signals.
2. Standardize gene expression levels across all tissue samples using StandardScaler.
3. Analyze covariance structures and calculate high-dimensional eigenvalue spectrum.

Part B: Dimensionality Reduction Pipelines

1. Apply Principal Component Analysis (PCA) and plot the Cumulative Explained Variance curve.
2. Apply t-SNE and Locally Linear Embedding (LLE) to project features into 2D and 3D manifold spaces.
3. Train a Random Forest or Support Vector Machine classifier on raw vs PCA-transformed features.

Part C: Evaluation, Visualization & Biological Inference

1. Construct 2D scatter plots of PCA, t-SNE, and LLE colored by tumor tissue categories.
2. Compare classification accuracy, training run-times, and feature dimensionality reduction ratios.
3. Discuss information loss versus computational speedup in biomedical machine learning pipelines.

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Question 8: Cross-Validation Strategies & Performance Evaluation for Solar Energy Forecasting

Dataset Reference: Solar Power Generation Dataset

(https://www.kaggle.com/datasets/anikannal/solar-power-generation-data)

Industry Case Scenario: A renewable energy provider needs an accurate regression pipeline to forecast daily solar power generation based on weather parameters, irradiation levels, and ambient temperature.

Part A: Exploratory Data Analysis & Temporal Profiling

1. Parse timestamps, extract diurnal temporal features, and identify missing power output entries.
2. Compute correlation heatmaps between ambient temperature, module temperature, irradiation, and DC power output.
3. Generate rolling average plots to examine daily generation seasonality.

Part B: Feature Engineering & Time-Series Cross-Validation

1. Engineer lag features (e.g., t-1 hour irradiation) and rolling statistics for generation values.
2. Implement TimeSeriesSplit (Rolling Window CV) and standard K-Fold CV to evaluate linear and tree-based regressors.
3. Perform hyperparameter tuning on Gradient Boosting Regressor using TimeSeriesSplit.

Part C: Model Evaluation, Error Analysis & Operational Inference

1. Calculate MAE, RMSE, MAPE, and R-squared metrics for validation folds.
2. Plot actual vs. predicted power output time series along with residual distribution plots.
3. Explain why standard K-Fold CV creates data leakage in time-series settings compared to TimeSeriesSplit.

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Question 9: Reinforcement Learning Control Optimization for Autonomous Lander Systems

Dataset Reference: Gymnasium / OpenAI Gym - LunarLander-v2 & MountainCar-v0

(https://gymnasium.farama.org/environments/box2d/lunar_lander/)

Industry Case Scenario: An aerospace robotics research center wants to benchmark reinforcement learning algorithms to safely control an autonomous module landing on a designated pad with minimal fuel consumption.

Part A: Environment Exploration & Baseline Agent

1. Initialize the Gym environment, inspect continuous state dimensions and discrete action controls.
2. Execute a random control policy for 100 episodes to record baseline reward statistics.
3. Plot state trajectories (position, velocity, angle) and analyze environment reward structures.

Part B: RL Agent Algorithm Implementation

1. Implement a Value-Based RL Agent (Deep Q-Network - DQN) with replay buffer and target network.
2. Implement a Policy-Based RL Agent (REINFORCE / Policy Gradient Method).
3. Configure hyperparameter schedules for discount factor (gamma), learning rate, and epsilon decay.

Part C: Performance Evaluation, Visualization & Control Inference

1. Plot learning curves (100-episode moving average reward vs. training steps) for both agents.
2. Compare convergence stability, sample efficiency, and landing success rate.
3. Provide engineering inferences on policy robustness against state perturbation during touchdown.

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Question 10: Group-Based Cross Validation for Multi-Class Cardiovascular Disease Risk Profiling

Dataset Reference: Cardiovascular Disease Dataset

(https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset)

Industry Case Scenario: A clinical research consortium requires a multi-class classification pipeline to stratify patient cardiovascular risk profiles based on objective physiological data, medical examination scores, and subjective lifestyle factors.

Part A: Exploratory Data Analysis & Patient Profiling

1. Examine target distribution and detect physiological anomalies (e.g., negative blood pressure values).
2. Perform univariate analysis on age (converted to years), systolic/diastolic blood pressure, cholesterol levels, and BMI.
3. Generate boxplots and violin plots comparing blood pressure ranges across risk outcome categories.

Part B: Feature Engineering & Group-Based Model Fitting

1. Engineer clinical risk metrics (e.g., Pulse Pressure = Systolic - Diastolic, BMI categories).
2. Fit Multi-Class Random Forest, Support Vector Machines (SVM), and XGBoost classifiers.
3. Implement GroupKFold Cross-Validation grouped by clinic location/patient group to prevent data leakage.

Part C: Model Evaluation, Diagnostics & Medical Inference

1. Compute One-vs-Rest ROC-AUC, Macro Precision, Recall, Macro F1-score, and Multi-Class Confusion Matrix.
2. Plot Multi-Class ROC Curves and Feature Importance bar graphs.
3. Formulate clinical inferences on top risk factors to assist early preventive medical interventions.

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